

Haoming Wang

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Research Profile

PhD candidate (ECE, University of Pittsburgh) working on **multimodal generative models**, **video benchmark synthesis**, and **latent-representation shaping**. **First-authored CVPR'26 Oral** on a procedural **photo-realistic 3D video benchmark generator** that controllably renders egocentric scenes and establishes new evaluation protocols for spatiotemporal reasoning under compositional, relational, and observational complexity. Direct experience with **Stable Diffusion bilevel mask learning** (FreezeAsGuard), **training-free decomposition of diffusion / GAN / autoregressive image generators** (FineXL), **Dirichlet-energy latent regularization of VLMs**, and **cross-frame consistency under sequential observation budgets** (MosaicThinker) – all immediately relevant to scalable long-range video generation, latent-structure design, drift mitigation, and perceptual evaluation.

Education

Ph.D., Electrical and Computer Engineering, University of Pittsburgh Sept 2022 – May 2027
(Anticipated)

Advisor: Prof. Wei Gao, Intelligent System Lab. Focus: multimodal generative / VL models, spatial reasoning, video benchmarks.

B.Eng., Automation, Zhejiang University GPA 3.8/4.0
Chu Kochen Honors College. Sept 2018 – May 2022

Selected Publications – Video Synthesis, Diffusion Models & Latent Representations

[CVPR'26 Oral] InfiniBench: Infinite Benchmarking for Visual Spatial Reasoning with Customizable Scene Complexity CVPR 2026

Haoming Wang, Qiyao Xue, Wei Gao. First author. Acceptance 25.4%; selected for Oral.
[\[arXiv:2511.18200\]](#) [\[Code\]](#) [\[Data\]](#)

- LLM-agent (Gemini-2.5-Pro) translates natural-language scene specifications into procedural constraints and refines them via a CoT feedback loop driven by BEV collision maps – a controllable pipeline for synthesizing **photo-realistic egocentric videos** (Blender Cycles) with parameterized compositional, relational, and observational complexity.
- Cluster-based layout optimizer (introducing a movable-cluster action space) and a **frontier-based task-aware camera trajectory planner** (Dijkstra over candidate viewpoints scored by validity / FoV / occlusion) generate consistent multi-second video sequences with controllable identity and layout – directly transferable to long-range video generation evaluation.
- Establishes a new **evaluation protocol** for video-based VLM reasoning across measurement / perspective-taking / spatiotemporal task families; outperforms LLM-based generators (I-Design, Holodeck, LayoutGPT) and procedural frameworks (Infinigen, Luminous) on prompt fidelity, CLIP alignment, realism, and physical plausibility – the only method avoiding the fidelity-vs-plausibility tradeoff at high complexity.

Deciphering Personalization: Towards Fine-Grained Explainability in Natural Language for Personalized Image Generation Models (FineXL) 2025 (Under Review)

Haoming Wang, Wei Gao. First author. [\[Paper\]](#)

Training-free decomposition of CLIP-space divergence between base and personalized **SD v2.1 / v3.5**, **ControlGAN**, **Anole-7B** autoregressive generators into orthogonal VLM-discovered concepts; cuts single-aspect MAE **56%**, multi-aspect selection error $\geq 50\%$; introduces a mixed-proportion methodology for fine-grained perceptual evaluation of generative-model differences.

FreezeAsGuard: Mitigating Illegal Adaptation of Diffusion Models via Selective Tensor Freezing 2024 (Preprint)

Kai Huang, *Haoming Wang* (co-author), Wei Gao. [\[Paper\]](#) [\[Code\]](#)

Bilevel optimization over Stable Diffusion tensors via sigmoid mask relaxation – a differentiable recipe for fine-grained manipulation of diffusion weights at scale; 37–47% stronger mitigation than UCE / IMMA on portrait / artwork / NSFW; 48% GPU memory and 21% wall-clock savings.

Uncovering and Shaping the Latent Representation of 3D Scene Topology in Vision-Language Models 2026 (Preprint)

Haoming Wang, Wei Gao. *First author*. [\[Paper\]](#) [\[Code\]](#)

Identify a VLM latent subspace aligned with **Laplacian eigenmaps** of the 3D scene graph; a **Dirichlet-energy regularizer** shapes this subspace at training, beating SFT by up to **12.1%** – a concrete instance of shaping hierarchical latent structure directly relevant to extended-generation latent design.

Cross-Frame Video & Multi-View Reasoning

MosaicThinker: On-Device Visual Spatial Reasoning for Embodied AI via Iterative Construction of Space Representation 2026 (Under Review)

Haoming Wang, Qiyao Xue, Weichen Liu, Wei Gao. *Co-first author*. [\[arXiv:2602.07082\]](#)

Operates on **RGB egocentric video**; **topology-aware MST cross-frame alignment** mitigates drift in sequential ICP-style chaining; **Gaussian-kernel iterative key-frame selection** as a sequential observation-budget policy (sister problem to chunked / autoregressive generation). Up to **40%** accuracy gain on cross-frame VSI-Bench / Metro-Spatial-QA.

ReMindView: Reasoning Path and Latent State Analysis for Multi-view Visual Spatial Reasoning 2026 (Under Review)

Qiyao Xue, Haoming Wang (co-author), Weichen Liu, Shiqi Wang, Yuyang Wu, Wei Gao. [\[arXiv:2512.02340\]](#) [\[Code\]](#)

Cognitively grounded multi-view benchmark (>50K VQAs); **linear-probe + entropy-dynamics** analysis on per-phase token logits separates correct vs. incorrect reasoning trajectories – a candidate perceptual / coherence metric for assessing temporal consistency in generated video.

Other First-Author Publications

[MobiSys'25] Never Start from Scratch: Expediting On-Device LLM Personalization via Explainable Model Selection MobiSys 2025

Haoming Wang, Boyuan Yang, Xiangyu Yin, Wei Gao. (18.0% acceptance.) [\[Paper\]](#)

[MobiCom'25] When Device Delays Meet Data Heterogeneity in Federated AIoT Applications MobiCom 2025

Haoming Wang, Wei Gao. (17.1% acceptance.) [\[Paper\]](#)

[AAAI'25] Tackling Intertwined Data and Device Heterogeneities in Federated Learning with Unlimited Staleness AAAI 2025

Haoming Wang, Wei Gao. (23.4% acceptance.) [\[arXiv\]](#)

Co-authored: *Spatial Reasoning Survey for VLMs (under review)*; *HEVEN, GlobalNav – VLM-based embodied navigation systems (3rd author, under review)*.

Research Experience

Graduate Student Researcher, Intelligent System Lab, University of Pittsburgh Sept 2022 – Present

- **(May 2025 – Present)** Procedural / LLM-agentic 3D video benchmark synthesis (CVPR'26 Oral); Dirichlet-energy latent shaping of VLM 3D-topology subspaces; cross-frame consistency under sequential observation budgets.
- **(Nov 2024 – May 2025)** Diffusion / generative-model interpretability and personalization: training-free fine-grained natural-language explanations of Stable Diffusion / GAN / autoregressive image generators (FineXL); on-device LLM personalization (Xpert).
- **(Sept 2022 – Oct 2024)** Federated learning systems with intertwined data and device heterogeneity (MobiCom'25, AAAI'25); large-scale distributed training pipelines.

Awards & Recognition

ECE Research Assistant of the Year, University of Pittsburgh April 2026
Department of Electrical and Computer Engineering; recognized at the EGSO event.

Technical Skills

Generative Models: Stable Diffusion, ControlGAN, Anole-7B, CLIP variants; bilevel mask optimization, linear-representation decomposition, Laplacian-eigenmap latent probing.

Video / 3D Pipelines: Blender Cycles, Infinigen, Habitat, Isaac Sim, frontier-based camera planning, MatchAnything, DepthPro, SAM-3D / 3DGS.

Programming: Python, PyTorch, CUDA, Hugging Face, ONNX, TensorRT.